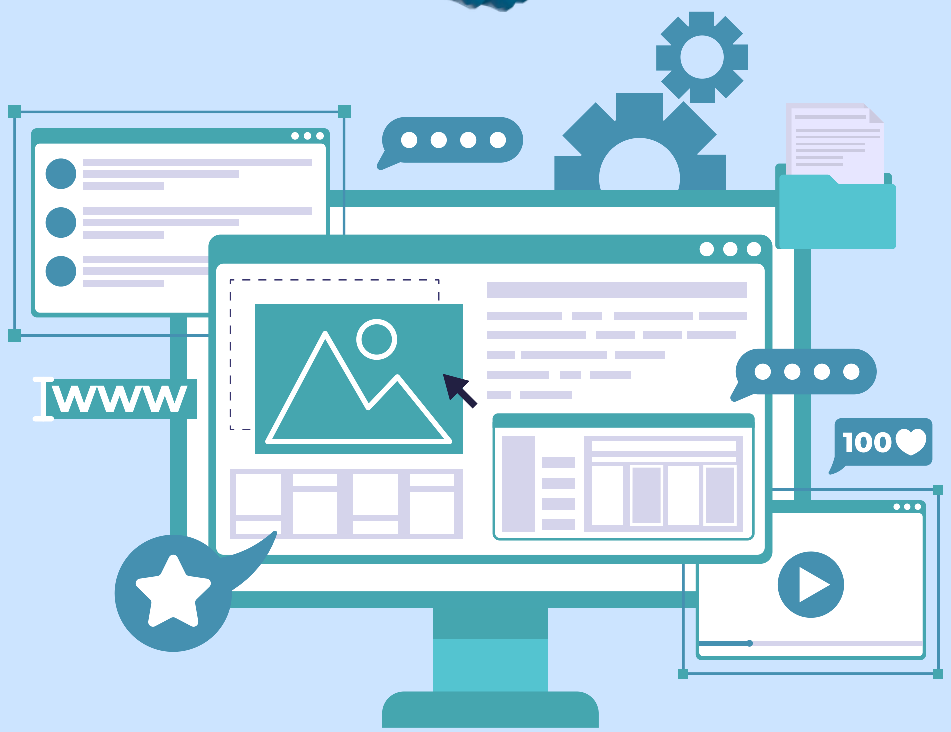
CSE602 PROJECT



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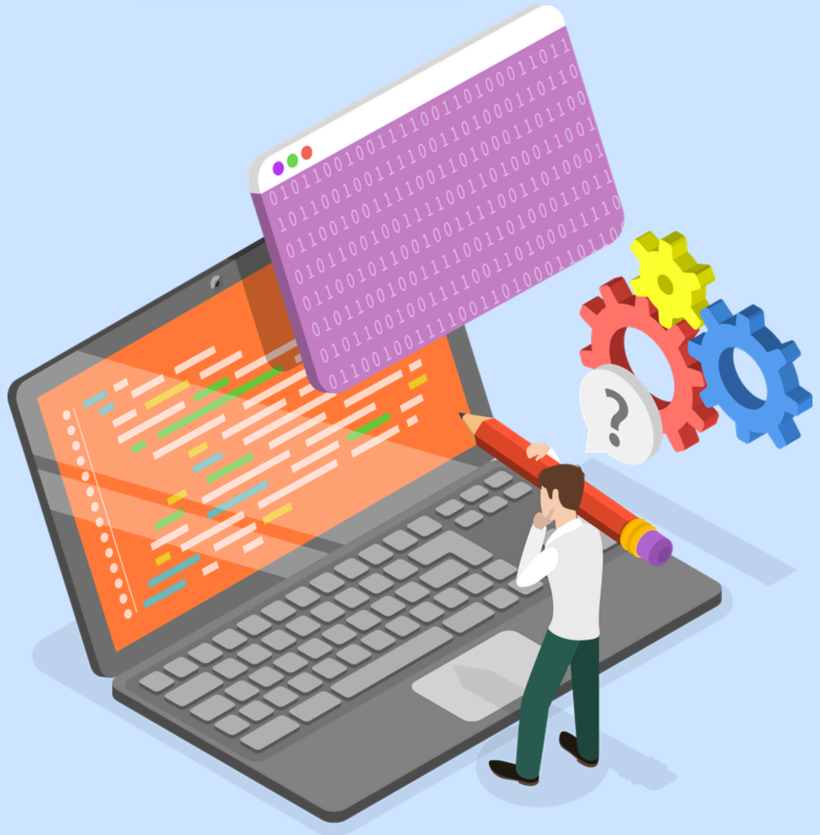
PROBLEM STATEMENT



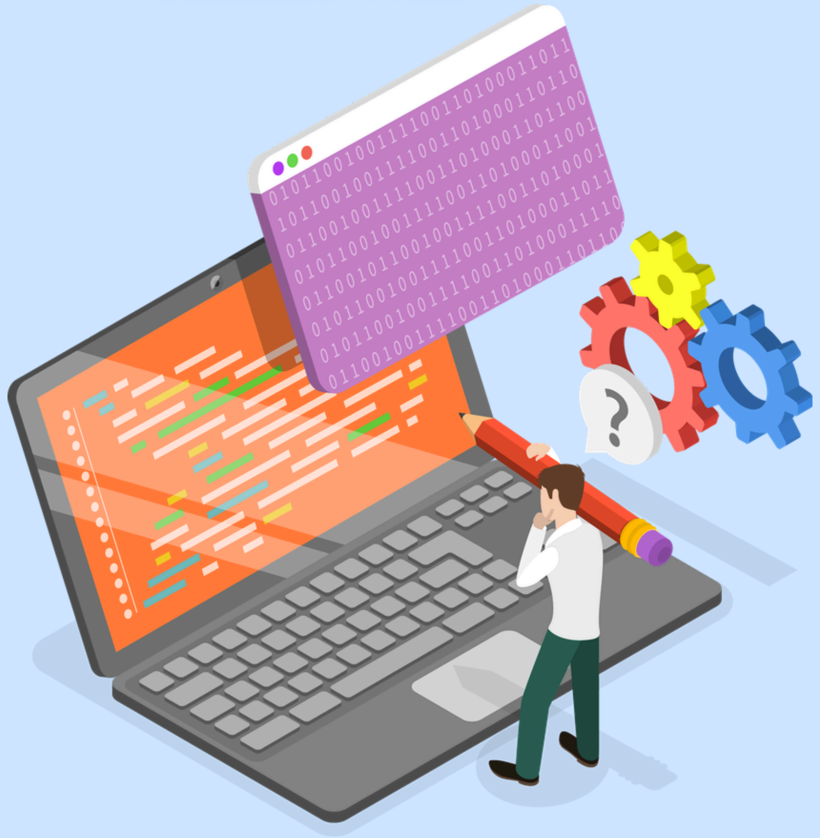
- Derived from the BONBID-HIE challenge listed on the Grand Challenges website.

- Diagnosis of Hypoxic Ischaemic Encephalopathy (HIE) in infants by using image segmentation to identify brain lesions (areas of structural damage) in the MRI images of infant brains.

- Task: Image Segmentation

SIGNIFICANCE

- The mortality rate from HIE can be as high as 60%.
- Survivors often experience significant cognitive impairment due to permanent brain damage.
- Effective diagnosis could result in better prognosis and timely treatment.
- Effective segmentation remains challenging as HIE brain lesions are diffuse or ‘multi-focal’ (scattered throughout the brain).



DATA DESCRIPTION

TRAINING DATA

- MRI scans of 133 HIE patients between 0 and 14 days old from the Massachusetts General Hospital.

  Dimensions = 128 x 128 x 23

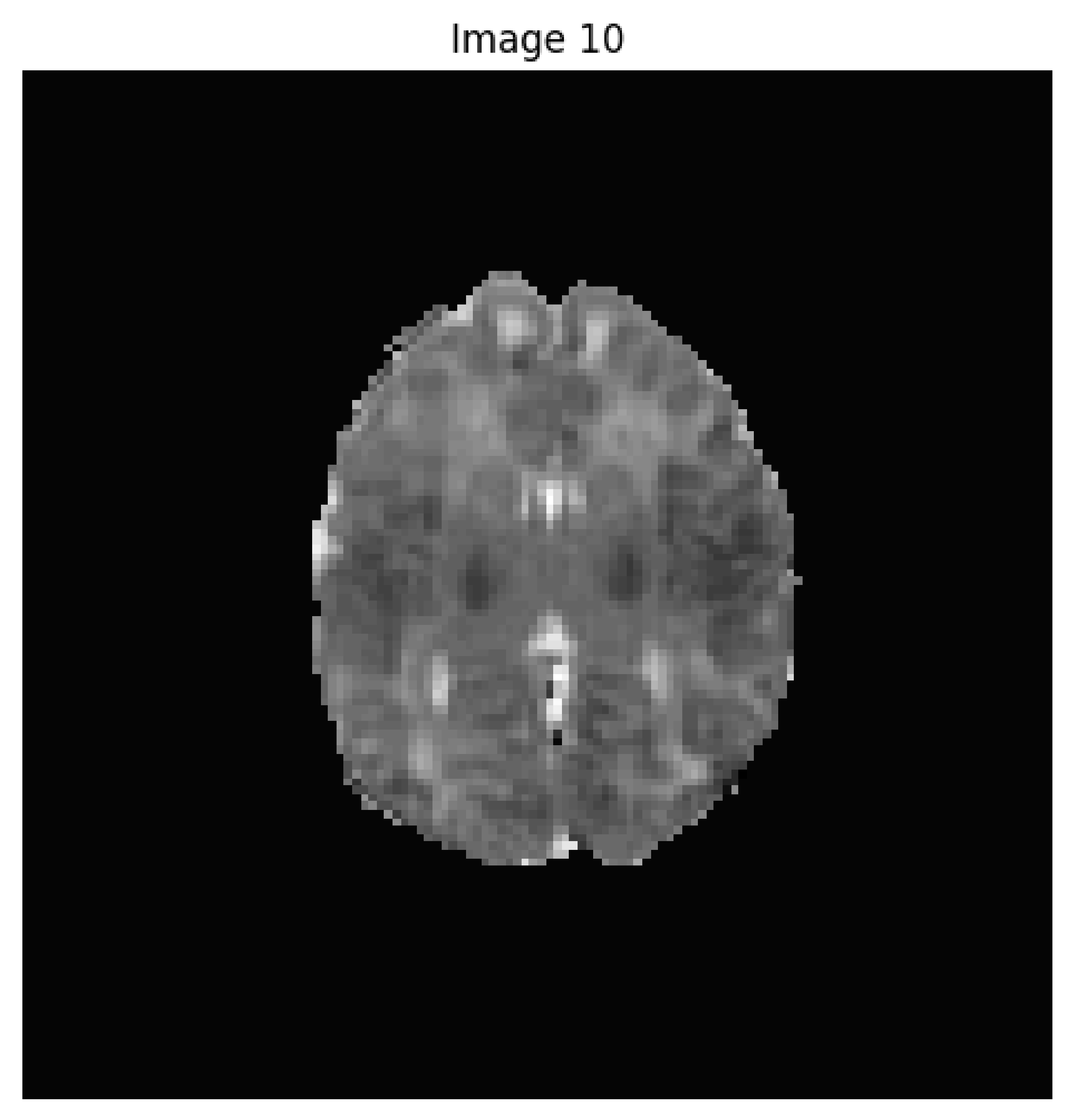
- A special type of MRI image known as Apparent Diffusion Coefficient (ADC) maps.

- ADC maps show the amount of fluid molecules (blood or water) diffusing across a region of the brain. This diffusion can be obstructed by HIE brain lesions.

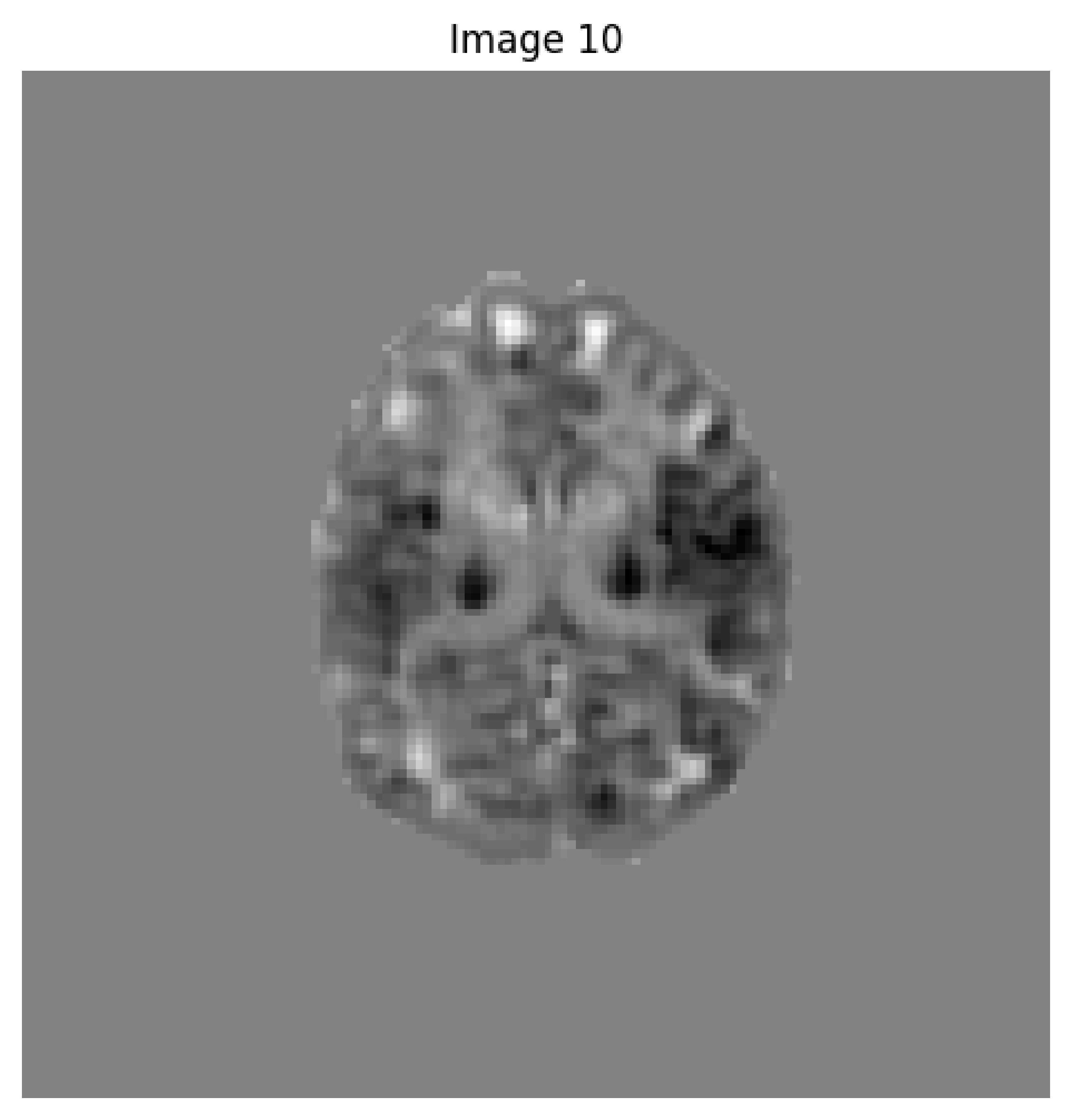
- Three types of images: raw images, preprocessed images with z-score normalisation and expert-annotated lesion masks.

RAW IMAGES

- ADC maps are used in clinical practice to diagnose HIE.
- Damaged brain regions (brain lesions) can restrict water diffusion or blood supply across the brain.
- Such areas of low diffusion appear darker than the surrounding brain regions.
- Doctors use these dark-coloured areas to diagnose HIE.



NORMALISED IMAGES

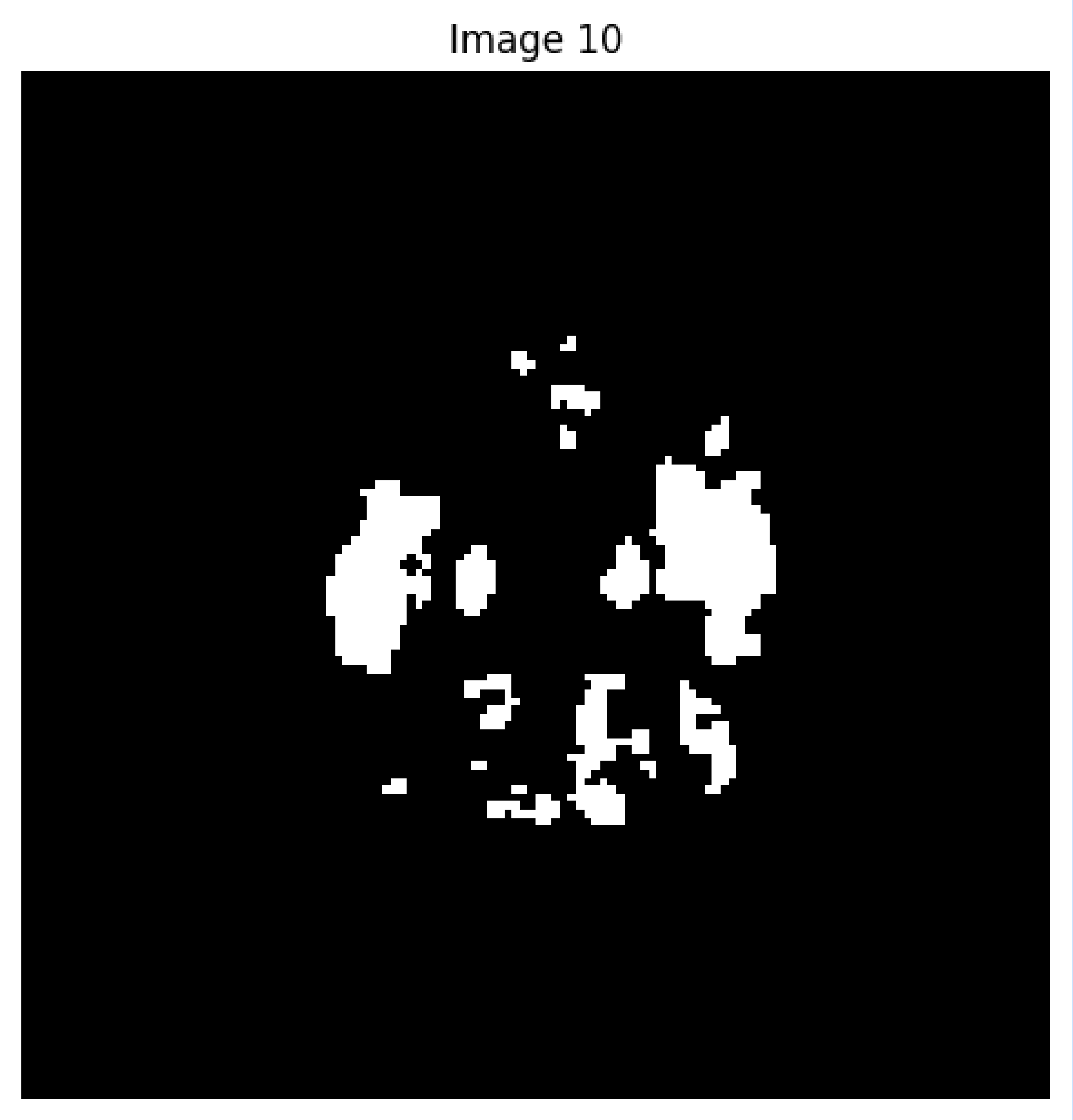


- Different brain regions have different healthy baselines for water or blood diffusion.
- Thus, what might be considered low diffusion for one part of the brain may be the normal amount for a different brain region.
- Z-score normalisation was used to account for these differences.
- The diffusion in each brain region is compared with the baseline for that specific region.

EXPERT ANNOTATED LESIONS

- Binary masks of the ADC images, depicting the location of HIE brain lesions.

- These lesions have been manually annotated by trained physicians with 3 or more years of experience.



MODEL ARCHITECTURE

- TransUNet: Transformer-enhanced UNet
- A hybrid model comprising a CNN (UNet) and a transformer.
- The transformer enables the model to capture long-term dependencies.
- This makes it especially useful for segmenting diffuse, small, scattered brain lesions.
- Potential Limitations: High computational cost and high amount of training data required

MODEL EVALUATION METRICS

- Dice Similarity Coefficient (DSC): Measures the spatial overlap between a predicted segmentation and the ground truth. Ranges from 0 (no overlap) to 1 (perfect match).

- Hausdorff Distance: Measures the worst-case difference between the predicted segment boundaries and ground truth boundaries. Ranges from 0 (no distance) to infinity.

REFERENCES

- BOston Neonatal Brain Injury Data
- Bao et al., 2023
- Radiopaedia - Apparent Diffusion Coefficient
- Soh et al., 2023
- Babbo et al., 2024
- Evaluation Metrics for Image Segmentation Models

THANK YOU